

Conditional Assembly of Gravity, Matter, and Abelian Gauge Terminals

from One Ordered Tau Descent

Tau Core Foundation Paper VII-A

Jozsef Olcsak

August 2026

Abstract

We ask whether the previously constructed Tau gravity, temporal, quantum, matter and Abelian gauge terminals can descend from one ordered physical source ledger rather than from independently fitted sector models. On a stabilized morphological body we construct one once-counted post-body effective functional whose effect, metric, gauge and clock-character derivatives generate probabilities, stress, current and formal clock response. Diffeomorphism and $U(1)$ gauge invariance yield the conditional Einstein–Maxwell–matter equations, Ward/Bianchi identities and their standard weak-field leaf. A disjoint occupation ledger prevents duplicate classical/quantum stress, while exact countermodels show that standard-limit agreement does not establish common Tau provenance.

On an occupied two-mode restriction, ROOT one-particle coherence and transverse-traceless stress arise from one preparation without an additional cross-terminal gain. Differential recovery is reduced to a source-rank problem: after de-duplication, the ROOT increment over the common P1 row space is fixed by one source-owned mixed Gram. The paper therefore gives a conditional common-source assembly and a precise non-identifiability boundary, not a first-principles derivation of Einstein–Hilbert or Maxwell action shapes, not quantized geometry and not unrestricted physical parent realization. The rank-seven P3 packet and scalar-15/RG specialization are now owned by Technical Papers VII-B and VII-C respectively.

Inputs from earlier papers. Paper I supplies body-first descent; Paper II typed finite support; Paper III recovery-correctable record transport; Paper IV the parent-realization boundary; Paper V observer time; Paper VI the conditional quantum terminal.

Results owned here. Morphology-conditioned joint-terminal assembly, the finite occupation criterion, one effective source, the disjoint stress ledger, Ward/Bianchi consistency, standard-form classification inside the declared leaf, ROOT–TT common preparation, Abelian directional activation and the exact de-duplicated source-rank obstruction.

Claim boundary. The Einstein–Hilbert and Maxwell action shapes are inputs of the declared standard Lorentzian leaf and are not derived from the narrow Tau parent. The results below are conditional assembly theorems. They do not prove that the unrestricted physical base-seed law realizes or occupies the common packet, quantize the metric, derive non-Abelian gauge occupation or reproduce the Standard-Model spectrum.

Notation and packet map

Symbol or packet	Role
$P1$	Gravity–time–radiation/matter differential packet.
ROOT	Occupied two-mode quantum one-particle descriptor; its normalized shape is ρ_{ROOT} and its occupation magnitude is retained separately.
TT	Transverse–traceless metric/stress readout paired with ROOT.
$P3$	Rank-seven primitive algebra/recovery packet, imported only at the handoff from Technical Paper VII-B.
$J_{\text{phys}}^{\text{lin}}$	Parent-side physical incidence on the source-frozen linear quotient.
K_*	Positive source metric/Hessian on the true-null quotient.

1 Question and reading order

The paper owns the conditional arrow

$$M_\tau \longrightarrow \text{one post-body source ledger} \longrightarrow \{g, A, \Phi, \rho_O, \text{formal clock response}\}.$$

It does not use terminal residuals to select M_τ and does not construct the body from a readout. The proof order is source ownership, variational derivatives, standard-limit controls, ROOT–TT common preparation and finally source-rank non-identifiability.

Contents

1	Question and reading order	2
2	Inherited-results ledger	4
3	Morphology-conditioned assembly and occupation	4
4	The ordered Paper-VII ledger	6
5	One effective source on each stationary body fiber	6
6	Disjoint occupation and stress ownership	8
7	Joint terminal theorem	9
8	Relation to established frameworks	10
9	Local constraint structure	10
10	Primitive terminal inventory	11
11	Why the standard limit does not prove common provenance	11
12	Quantum backreaction and the quantum-gravity boundary	12
12.1	Two-mode common-preparation theorem	13
12.2	Minimal invariant activation potential	21
12.3	Source-loaded activation from an occupied Abelian field	21

12.4 Faithful common-completion occupation	23
12.5 Exact source-rank reduction	24
12.6 De-duplicated primitive recovery dimension	25
13 Technical-paper handoffs	26
14 Decision ledger	27
15 Occupied-body input inherited from parent realization	27
16 Conclusion	27
A Self-contained dependency certificate	27

2 Inherited-results ledger

Result	Owner	Status and assumptions	Use here
Body-first descent	Paper I	conditional physical MVP	generative order
Typed incidence	Paper II	finite source theorem	occupied domains
Record recovery	Paper III	exact/approximate conditional theorem	observer terminal
Parent-law realization	Paper IV	present reduct does not entail full packet	claim boundary
Temporal descent	Paper V	conditional clock/causal leaf	time and weak limit
Quantum terminal	Paper VI	conditional complex/effect/QFT packet	Γ_q and ρ_O
Morphology-conditioned readouts	theory hub	common-source assembly for an occupied complete signature	typed terminal projections
Operation-generated occupation	theory hub	cyclic-surjective/root-positive criterion inside a fixed source representation	block nonvanishing
P1 assembly	theory hub	conditional metric/time/radiation ledger	g , null radiation, stress
P3 Abel handoff	theory hub	conditional $Z_F = p_* K_Y$	Maxwell/current sector
Pointed P3 exterior seed	theory hub	conditional orthonormal 256-vector root orbit; full-rank nonracial countermodel	representation boundary
Support-port P3 synthesis	theory hub	commuting support flag, two ordered ports, independent unit and smooth decoder; finite audit	rank-seven source reconstruction
Tau-local composition origin	theory hub; re-proved here	exact inside the component-local no-bridge/no-hidden-global-relation class	two-leg record and effect tensor algebra
Component-local P3 joint occupation	owned here	exact inside one once-counted enriched common law; narrow-reduct selection disproved	joint LC1–LC4 and rank-seven status
Scalar-15/type-II completion	theory hub branch	P4 class-relative base-seed reconstruction, common-scale ratio and exact functional RG law; unrestricted branch occupation open	downstream particle-sector handoff

Table 1: Dependency and ownership ledger. No inherited conditional result is promoted to an unrestricted physical parent-law theorem.

For logical self-containment, Appendix A restates every imported premise actually used below. The paper labels in Table 1 are provenance locators, not hidden hypotheses: the conditional arguments can be checked from the restated premises without accepting an unquoted result from a companion manuscript. Several listed papers provide architectural context only and are not logical dependencies of the main theorems.

3 Morphology-conditioned assembly and occupation

The inherited terminals are not treated as independent information channels. Let the stabilized body and observer support determine one finite physical source representation $\mathcal{E}_{*,O}$. No direct-sum

decomposition of this source is assumed. Instead, for the typed role set

$$\mathcal{P} = \{T, G, R, M, Q, A\}, \quad \mathcal{Y}_{\text{role}} := \bigoplus_{a \in \mathcal{P}} \mathcal{Y}_a, \quad (1)$$

let $R_a : \mathcal{E}_{*,O} \rightarrow \mathcal{Y}_a$ be terminal-role maps and $R = (R_a)_a : \mathcal{E}_{*,O} \rightarrow \mathcal{Y}_{\text{role}}$ their stacked map. The direct sum is therefore a de-duplicated *role codomain*, not a claim that the source information separates. Common source directions may affect several R_a simultaneously; their overlap is measured by the joint kernel and stacked rank of the R_a .

Let A_O be the body- and observer-conditioned access map and let $Q_O^{(a)}$ be a stable observer-relative resolution map. The operational terminals are

$$Y_O^{(a)} = Q_O^{(a)} R_a A_O \Xi_*, \quad (2)$$

for one occupied source representative Ξ_* . Equality under the specified $Q_O^{(a)}$, rather than a non-transitive distance threshold, defines the resolved readout classes. Thus observer-relative quantization remains part of the full descent and is not replaced by the continuous source geometry.

Proposition 1 (Conditional morphology-conditioned representation). *Assume a stabilized body, one occupied complete source representation $\mathcal{E}_{*,O}$, a source-owned typed ledger map, and stable resolution maps $Q_O^{(a)}$. Then all terminals in Eq. (2) are functorial restrictions of one source representative. Their common-source ownership and once-counting follow without terminal-specific gains. Different terminal names do not by themselves imply independent information. This is a representation theorem for an already occupied complete source ledger; it does not establish uniqueness among all admissible source functionals.*

Proof. Body stabilization fixes the domain of A_O . The single map $A_O \Xi_*$ is then mapped by the typed R_a and resolved only afterwards by $Q_O^{(a)}$. Consequently every terminal factors through the same occupied representative and the same access map. Any additional terminal gain would be an extra source datum, contrary to the assumed complete ledger. The conclusion is role-completeness of this construction, not uniqueness under field redefinitions, improvement terms or alternative ultraviolet completions. $\square$

Let $\mathfrak{A}_{\text{src}}$ be the source-owned operation algebra and Ω_s the occupied source vector, and define $V_s := \text{span}(\mathfrak{A}_{\text{src}} \Omega_s)$.

Definition 1 (Activation and full occupation). *A required terminal role a is activated when $R_a(V_s) \neq \{0\}$. It is fully occupied when*

$$R_a(V_s) = \mathcal{Y}_a. \quad (3)$$

Full occupation implies activation; the converse need not hold.

Assembly does not by itself prove either property. In finite dimension, full occupation of every required role is equivalent to

$$R_a(\text{span}(\mathfrak{A}_{\text{src}} \Omega_s)) = \mathcal{Y}_a \quad \text{for every required } a. \quad (4)$$

Proposition 2 (Operation-generated occupation criterion). *If Ω_s is cyclic for $\mathfrak{A}_{\text{src}}$ on $\mathcal{E}_{*,O}$ and each required R_a is surjective, then Eq. (4) holds and every required role is fully occupied. More*

weakly, let $X_0 \geq 0$ be the occupied root covariance with $\text{supp } X_0 \subseteq V_s$, and let the source-owned path maps L_e preserve the occupied subspace, $L_e(V_s) \subseteq V_s$. Define

$$X_a := \sum_{e \in \mathcal{P}_a} R_a L_e X_0 L_e^\dagger R_a^\dagger \geq 0. \quad (5)$$

If $R_a L_e X_0^{1/2} \neq 0$ for at least one $e \in \mathcal{P}_a$, then role a is activated, but full rank is not implied.

Proof. Cyclicity gives $V_s = \mathcal{E}_{*,O}$; surjectivity then gives $R_a(V_s) = \mathcal{Y}_a$. In the weaker case, every summand in Eq. (5) is positive semidefinite and

$$\text{Tr } X_a = \sum_{e \in \mathcal{P}_a} \|R_a L_e X_0^{1/2}\|_{\text{HS}}^2. \quad (6)$$

One nonzero path therefore implies $X_a \neq 0$. Because $\text{Ran}(L_e X_0^{1/2}) \subseteq V_s$, its nonzero image lies in $R_a(V_s)$ and hence proves activation. It does not establish Eq. (3). $\square$

Propositions 1 and 2 move the open problem upward. Once the common representation and its source operation law are occupied, separate terminal selection is unnecessary. What remains unproved is that $B_\tau^{\text{phys}} + s_U$ selects this representation, makes Ω_s cyclic with surjective required role maps (or realizes the rooted positive alternative), and chooses the observer access and resolution maps. Algebraic generation, terminal agreement and post-handoff record completeness do not prove that prior physical realization and occupation.

4 The ordered Paper-VII ledger

Let y denote body coordinates, and write $z = (g, A, \Phi)$ for the variational post-body geometry, Abelian gauge and occupied classical matter fields. The quantum preparation is instead a fixed positive trace-one boundary datum $\rho_{\text{in},O} \geq 0$, $\text{Tr } \rho_{\text{in},O} = 1$ for the in-in contour. It is not an unconstrained stationary field.

Definition 2 (Once-counted ordered ledger). *The Paper-VII ledger is*

$$\mathfrak{L}_7 = (\mathcal{A}_{\text{pre}}[y; B, s], y_*, \Gamma_7[g, A, \Phi \mid y_*; \rho_{\text{in},O}], \text{Desc}_{O \leftarrow y_*}, \mathcal{U}), \quad (7)$$

where $D_y \mathcal{A}_{\text{pre}}(y_*) = 0$, the physical body Hessian is positive, and

$$\Gamma_7 = \Gamma_{\text{grav}}[g \mid y_*] + \Gamma_{\text{EM}}[A, g \mid y_*] + \Gamma_{\text{occ}}[\Phi, A, g \mid y_*] + \Gamma_q[A, g \mid y_*; \rho_{\text{in},O}] + \Gamma_{\text{ct}}[A, g]. \quad (8)$$

Each occupied mode and each counterterm appears in exactly one summand. Observer restriction is evaluated after stationarity and is not varied back against y_* . The state $\rho_{\text{in},O}$ fixes the CTP preparation; the Euler equations vary g , A and Φ , not $\rho_{\text{in},O}$. The unit packet $\mathcal{U}$ is shared by every summand.

5 One effective source on each stationary body fiber

The symbol Γ_7 need not be postulated as five unrelated pieces. Let φ collect the occupied quantum fields, and for each declared outcome family let the dimensionless constant probe η_a couple to a

normalized POVM-effect insertion satisfying $E_a \geq 0$ and $\sum_a E_a = I$ (hence $0 \leq E_a \leq I$). Let ϑ denote the occupied clock character. The source convention is chosen so that

$$\Gamma_{\text{PC}}[\eta] = \Gamma_{\text{PC}}[0] + \sum_a \eta_a E_a[\varphi], \quad \langle E_a \rangle = \text{Tr}(\rho_O E_a). \quad (9)$$

On the already selected stationary body y_* define the doubled closed-time-path influence functional

$$\begin{aligned} \mathcal{I}_{\text{CTP}} := & \Gamma_{\text{PC}}[y_*; g^+, A^+, \vartheta^+, \eta^+, \varphi^+] \\ & - \Gamma_{\text{PC}}[y_*; g^-, A^-, \vartheta^-, \eta^-, \varphi^-], \end{aligned} \quad (10)$$

$$\exp\left(\frac{i}{\hbar} W_O[g^\pm, A^\pm, \vartheta^\pm, \eta^\pm \mid y_*; \rho_{\text{in},O}]\right) = \int_{\rho_{\text{in},O}} \mathcal{D}\varphi^+ \mathcal{D}\varphi^- \exp\left(\frac{i}{\hbar} \mathcal{I}_{\text{CTP}}\right). \quad (11)$$

The initial boundary condition joins the two branches through the fixed preparation $\rho_{\text{in},O}$, and the path integral includes the standard final branch identification. The in-in contour is required for real causal expectation values [Schwinger, 1961, Keldysh, 1965]. For every doubled source write $X_c = (X^+ + X^-)/2$ and $X_\Delta = X^+ - X^-$. All derivatives below mean $\delta/\delta X_\Delta$ followed by the physical limit $X_\Delta = 0$, $X_c = X$; differentiating an already identified common branch would instead lose the response by CTP normalization. The construction is a terminal representation of an atemporal ordered functional fibration, not a parent-time history. CTP evolution and observer restriction produce the normalized output state ρ_O used in Eq. (9); neither $\rho_{\text{in},O}$ nor ρ_O is varied as a local Euler field. Renormalization gives one W_O^{ren} whose physical-limit difference derivatives define

$$p_a = \left. \frac{\delta W_O^{\text{ren}}}{\delta \eta_a} \right|_{\eta=0} = \text{Tr}(\rho_O E_a) \in [0, 1], \quad \langle T_{\mu\nu}^q \rangle_{\text{ren}} = -\frac{2}{\sqrt{-g}} \frac{\delta W_O^{\text{ren}}}{\delta g^{\mu\nu}}, \quad (12)$$

$$\langle J_q^\mu \rangle_{\text{ren}} = -\frac{1}{\sqrt{-g}} \frac{\delta W_O^{\text{ren}}}{\delta A_\mu}, \quad \mathcal{R}_O^{\text{clock}} = \frac{\delta W_O^{\text{ren}}}{\delta \vartheta}. \quad (13)$$

Consequently $\sum_a p_a = 1$ within each normalized outcome family. Without the POVM and source-coupling conditions, the derivative would be only a generic operator expectation and could not be identified as a Born probability. Here the effect-source renormalization convention is required to preserve the POVM normalization at $\eta = 0$; Eq. (9) is an admitted operational insertion and not a claim that every POVM has a local polynomial field representation. The final expression is schematic until the clock character and its normalization are fixed, but its ownership is unambiguous. The minus sign in the current definition is fixed by the Maxwell convention $\Gamma_{\text{EM}} = -(Z_F/4) \int \sqrt{-g} F^2$: stationarity then gives $\nabla_\mu (Z_F F^{\mu\nu}) = J^\nu$. We use the same sign convention for the classical current, $J_{\text{cl}}^\mu := -(\sqrt{-g})^{-1} \delta \Gamma_{\text{occ}} / \delta A_\mu$.

Theorem 1 (Effective common-source construction). *Assume the ordered pre-body solve, a source-owned probe-complete post-body functional Γ_{PC} , an observer-admissible integration class $\mathcal{C}_O$ with fixed normalized boundary state $\rho_{\text{in},O}$, and a local covariant, anomaly-free renormalization prescription for the symmetries used below. Then Eqs. (11)–(13), together with the classical body and coherent mean-field terms, construct the Paper-VII Γ_7 ledger from one post-body source functional. The construction is complete at the level of the declared physical roles, modulo relocation of a finite local counterterm between the geometric and quantum bookkeeping columns while holding their sum fixed. No uniqueness claim is made across field redefinitions, improvement terms or inequivalent microscopic functionals.*

Proof. Functional differentiation of one W_O^{ren} gives the four displayed terminal owners. Diffeomorphism and gauge invariance give their common Ward covector. The fluctuation Green operator is the inverse of the corresponding gauge-fixed Hessian on the physical quotient (or its prescribed generalized inverse when zero modes remain) and therefore computes response; inserting it again as an action summand would duplicate the fluctuation contribution. A finite local counterterm can be written in either the geometric or renormalized quantum column, but linearity of variation shows that only the sum affects the field equations. Including the same term in both columns doubles its variation. Thus this assumed functional supplies a consistent role ledger, while its harmless counterterm placement convention is not fixed. The argument does not exclude a distinct probe-complete functional with the same terminal equations. $\square$

Scope. Theorem 1 is a construction theorem inside the selected ordered completion. It does not show that the physical base-seed law selects Γ_{PC} , $\mathcal{C}_O$, its field spectrum or its renormalized continuum measure.

The ordering is structural. If $H_B, H_P > 0$ and the post-body equations have body derivative C , the linear response is lower triangular:

$$\begin{pmatrix} \delta y \\ \delta z \end{pmatrix} = \begin{pmatrix} H_B^{-1} & 0 \\ -H_P^{-1}CH_B^{-1} & H_P^{-1} \end{pmatrix} \begin{pmatrix} \delta j_B \\ \delta j_P \end{pmatrix}. \quad (14)$$

Thus morphology conditions the physical terminals, while a later terminal load does not rebuild the body.

Proposition 3 (Simultaneous-action no-go). *For an invertible positive simultaneous Hessian*

$$\mathbb{H} = \begin{pmatrix} H_B & C^\dagger \\ C & H_P \end{pmatrix}, \quad (15)$$

exact vanishing of body response to every pure post-body load implies $C = 0$. Hence nonzero body conditioning and exact body freeze cannot both be encoded by one simultaneous unconstrained Hessian solve.

Proof. Schur inversion gives $\delta y = -H_B^{-1}C^\dagger(H_P - CH_B^{-1}C^\dagger)^{-1}\delta j_P$. If this vanishes for every δj_P , invertibility forces $C = 0$. $\square$

6 Disjoint occupation and stress ownership

Let the occupied mode labels form a disjoint union

$$\mathcal{I}_{\text{occ}} = \mathcal{I}_{\text{cl}} \sqcup \mathcal{I}_q. \quad (16)$$

Classical coherent modes represented in Γ_{occ} are excluded from the state-dependent part of Γ_q ; vacuum polarization and local renormalization terms occur only in $\Gamma_q + \Gamma_{\text{ct}}$. This disjoint partition is a sufficient transparent ledger, not the only possible field-theoretic organization. A background-field, one-particle-irreducible or two-particle-irreducible construction is also admissible when its mean field, fluctuation determinant and counterterms are derived from one effective action and no full contribution is inserted twice.

Proposition 4 (Duplicate-stress no-go). *If one nonzero occupied mode i is included in both $\mathcal{I}_{\text{cl}}$ and $\mathcal{I}_q$ with the same metric variation, its stress contribution is counted twice. No terminal redefinition removes this without changing the metric equation or adding a compensating subtraction.*

Proof. Metric variation is linear. The duplicated summands give $-2(\sqrt{-g})^{-1}\delta(\Gamma_i + \Gamma_i)/\delta g^{\mu\nu} = 2T_{\mu\nu}^{(i)}$. A subtraction merely moves the ownership entry and must therefore be recorded explicitly in Γ_{ct} . $\square$

7 Joint terminal theorem

Proposition 5 (Conditional standard-leaf form selector). *Assume that the selected readout leaf is a smooth four-dimensional Lorentzian manifold. Suppose the pure geometric action is local, diffeomorphism invariant, metric-only, and has metric Euler equations of at most second derivative order. Suppose also that the Abelian action is local, $U(1)$ -gauge invariant, parity even, quadratic in first derivatives of A , contains no derivatives of F , and has a constant positive kinetic coefficient. Then, up to boundary and four-dimensional topological terms, the two actions have the forms*

$$\Gamma_{\text{grav}} = \frac{1}{2\kappa} \int d^4x \sqrt{-g} (R - 2\Lambda), \quad (17)$$

$$\Gamma_{\text{EM}} = -\frac{Z_F}{4} \int d^4x \sqrt{-g} F_{\mu\nu} F^{\mu\nu}, \quad Z_F > 0. \quad (18)$$

The proposition fixes the action shapes inside this infrared class, but not κ , Λ , Z_F , or physical realization of the class itself.

Proof. The four-dimensional Lovelock classification leaves only the cosmological and Einstein tensors in the local metric-only second-order Euler tensor; the Gauss–Bonnet density is topological in four dimensions [Lovelock, 1971]. Integration gives the Einstein–Hilbert form up to boundary/topological terms and overall normalization. Abelian gauge invariance restricts a local first-derivative action to dependence on $F = dA$. In four dimensions the quadratic Lorentz scalars are $F_{\mu\nu} F^{\mu\nu}$ and $F_{\mu\nu} \tilde{F}^{\mu\nu}$. Parity removes the second from the dynamical class (and for constant coefficient it is topological), so the first is unique up to Z_F ; positivity fixes its sign. These classification arguments neither fix the coefficients nor prove that the parent selects their hypotheses. $\square$

The selector does not rule out higher-curvature gravity, torsion, additional geometric fields, nonlinear electrodynamics or parity-odd sectors; those lie outside its hypotheses. Its content is uniqueness within the declared standard infrared class, not uniqueness among all physical completions.

For the joint theorem adopt Proposition 5, the conditional common-source handoff $Z_F = p_* K_Y$, and a matter functional, quantum measure and renormalization prescription that are diffeomorphism and $U(1)$ gauge invariant and anomaly-free in those symmetries. The handoff removes an independent Maxwell gain inside the declared completion; it does not evaluate p_* or select the occupied representation trace K_Y from the physical parent.

Theorem 2 (Common-descent Einstein–Maxwell–matter assembly). *Under the ordered ledger, disjoint occupation rule, common unit packet and the stated invariances, stationarity of Eq. (8) with $\rho_{\text{in},0}$ held fixed gives*

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa \left(T_{\mu\nu}^{\text{EM}} + T_{\mu\nu}^{\text{cl}} + \langle T_{\mu\nu}^q \rangle_{\text{ren}} \right), \quad (19)$$

$$\nabla_\mu (Z_F F^{\mu\nu}) = J_{\text{cl}}^\nu + \langle J_q^\nu \rangle_{\text{ren}}. \quad (20)$$

On the matter and gauge equations, the Ward identities and Bianchi identity imply

$$\nabla^\mu T_{\mu\nu}^{\text{tot}} = 0, \quad \nabla_\mu J_{\text{tot}}^\mu = 0. \quad (21)$$

No additional gravity, gauge or quantum-backreaction gain is introduced at the terminal stage.

Proof. Equations (19) and (20) are the metric and gauge Euler equations of the single post-body functional. Gauge invariance gives current conservation. Diffeomorphism invariance gives the combined stress Ward identity, including the Lorentz-force exchange between matter and gauge sectors. The divergence of the left side of Eq. (19) vanishes identically, so the equations are mutually consistent. Disjoint occupation and the single-entry counterterm ledger ensure that each contribution is varied once. $\square$

Corollary 1 (Standard weak-field leaf). *For $g_{00} = -(1 + 2\Phi_N/c^2)$, stationary weak fields, nonrelativistic sources and a local patch whose boundary data admit the Newtonian limit, Eq. (19) reduces to*

$$\nabla^2 \Phi_N = 4\pi G \rho_{\text{tot}} - \Lambda c^2, \quad (22)$$

where ρ_{tot} is the same once-counted density appearing in the ledger. On scales $L \ll |\Lambda|^{-1/2}$, or for $\Lambda = 0$, the final term is negligible and the ordinary Poisson equation follows. Equation (20) reduces to the ordinary Maxwell equations after the inherited field normalization. This is standard recovery, not a Tau-specific deviation.

8 Relation to established frameworks

The use of one renormalized effective action and its metric variation belongs to semiclassical gravity and curved-spacetime QFT, not uniquely to Tau Core [Wald, 1977, Birrell and Davies, 1982, Hollands and Wald, 2001]. The closed-time-path extension and stress-noise cumulants likewise parallel stochastic gravity [Hu and Verdaguer, 2008]. Observer-relative descriptions have independent precedent in quantum-reference-frame constructions [Giacomini et al., 2019], while deriving Einstein dynamics from more primitive thermodynamic data is exemplified by emergent-gravity programs [Jacobson, 1995].

The finite-dimensional algebraic ingredients are also established mathematics rather than Tau-specific discoveries. The exterior realization of Clifford multiplication is standard spin geometry [Lawson and Michelsohn, 1989]; cyclic representations and their unitary uniqueness belong to the GNS/operator-algebra framework [Bratteli and Robinson, 1979]; and special dagger–Frobenius structures provide the finite-dimensional algebraic language for normalized copy/record data [Vicary, 2011, Coecke et al., 2013]. Paper VII’s claim is therefore not novelty of these ingredients separately, but their conditional placement in one source-owned, once-counted terminal ledger and the accompanying non-identifiability controls.

Paper VII’s narrower Tau-specific contribution is the body-first ordering, the typed morphology-conditioned access and resolution maps, the explicit once-counted cross-terminal ownership ledger, and the finite no-go separating terminal agreement from common provenance. The standard effective-action machinery is used as a consistency leaf; it is not presented as evidence for the Tau parent ontology.

9 Local constraint structure

The covariant equations must reproduce constraints, not only evolution equations. Let n^μ be the unit normal to an observer-compatible spacelike leaf, $h_{\mu\nu} = g_{\mu\nu} + n_\mu n_\nu$ its induced metric, and K_{ij} its extrinsic curvature. With $\rho = n^\mu n^\nu T_{\mu\nu}^{\text{tot}}$ and $j_i = -h_i^\mu n^\nu T_{\mu\nu}^{\text{tot}}$, the normal projections of

Eq. (19) give, in the stated sign convention,

$${}^{(3)}R + K^2 - K_{ij}K^{ij} = 2\kappa\rho + 2\Lambda, \quad (23)$$

$$D_j(K^{ij} - h^{ij}K) = \kappa j^i. \quad (24)$$

The normal projection of Eq. (20) gives the Abelian Gauss constraint

$$D_i(Z_F E^i) = \rho_J. \quad (25)$$

The Ward identities propagate these constraints when the classical and quantum matter equations hold. Paper VII therefore imports the standard local constraint architecture from the one joint functional; it does not derive a new constraint algebra or quantize it.

10 Primitive terminal inventory

The assembled packet contains several typed outputs, not several names for the same information:

Family	Primitive terminal content	Derived or composite descendants
Geometry/gravity	g , connection, curvature and constraints	proper time, tidal response, lensing after a null source
Gauge/radiation	A , F , conserved current, two screen polarizations	photon propagation, redshift and optical transport
Matter occupation	occupied localized or propagating solutions Φ	mass, energy, stress and charge records
Quantum operation	ρ , effects, instruments and correctable composition	Born statistics and renormalized quantum stress/current
Observer access	causal orientation, clock and stable restriction	observer-relative terminal classes

Stress-energy, current, redshift and lensing are variational or composite outputs of these families; they are not silently counted as new primitive source directions. Conversely, the symmetric local gravity descriptor cannot stand in for internal gauge charge, quantum phase or occupied matter. A global information-completeness claim must wait for the non-Abelian and fermionic known-terminal completions.

11 Why the standard limit does not prove common provenance

Proposition 6 (Equal-terminal provenance no-go). *The equations (19)–(21) do not imply that their coefficients or sectors arise from one Tau source.*

Proof. Construct two source models with the identical numerical effective functional $\Gamma_7[g, A, \Phi]$. In C_{joint} its sector coefficients are images of one common source line and unit map. In C_{sep} take four independent source lines and choose their terminal coefficient maps to have the same numerical images as the four coefficients in C_{joint} . The two models therefore have identical functionals as functions of the terminal fields, so all Euler equations, Ward identities and solution sets coincide. Their source ownership graphs are nevertheless non-isomorphic: one has a common source line and the other has four independent lines. Hence no observation or algebraic manipulation of Eqs. (19)–(21) alone can recover common provenance. This is an identifiability counterexample, not a claim that the independently sourced model is physically preferred. $\square$

This no-go fixes the empirical interpretation: terminal data may test a source-frozen forward ledger, but may not choose its source partition, susceptibilities or unit map.

Proposition 7 (Same-reduct selection no-go). *The currently frozen stationary body, common action line, observer quotient and terminal type signatures do not entail physical occupation of the full effective-source packet of Theorem 1.*

Proof. Hold fixed the body y_* , its Hessian, common units, causal orientation and observer quotient. Add the same typed internal edge carrier $\mathcal{H}_e = \mathbb{C}^2$ to both completions, with effect

$$E_e = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad \rho_w = \begin{pmatrix} 1-w & 0 \\ 0 & w \end{pmatrix}, \quad 0 < w < 1, \quad c_0 = 0, \quad c_1 = 1. \quad (26)$$

ρ_w is positive, has unit trace and is invariant under the diagonal $U(1)$ action. A concrete covariant edge functional is supplied by a charged scalar doublet $\chi \in \mathbb{C}^2$ with

$$\Gamma_e[\chi, g, A] = - \int d^4x \sqrt{-g} \left[g^{\mu\nu} (D_\mu \chi)^\dagger D_\nu \chi + m_e^2 \chi^\dagger \chi \right], \quad D_\mu = \partial_\mu - iq A_\mu I_2. \quad (27)$$

It is diffeomorphism invariant, diagonal- $U(1)$ gauge invariant and has the same positive internal state normalization in both completions. Define

$$\Gamma_{\text{PC}}^{(j)}[\eta_e] = \Gamma_{\text{common}} + \Gamma_e + c_j \eta_e \mathbf{E}_e, \quad j \in \{0, 1\}. \quad (28)$$

At the physical source value $\eta_e = 0$, both completions have exactly the same action, state, body, Hessian, observer quotient, typed carrier, covariance and normalization. The frozen upstream reduct does not include a law selecting the probe coefficient c_j . Nevertheless,

$$\left. \frac{\delta W_O^{(0)}}{\delta \eta_{e,\Delta}} \right|_{\text{phys}} = 0, \quad \left. \frac{\delta W_O^{(1)}}{\delta \eta_{e,\Delta}} \right|_{\text{phys}} = \text{Tr}(\rho_w E_e) = w > 0. \quad (29)$$

Only C_1 contains the declared nonzero effect-source owner. Since the full effective-source packet includes that owner, the frozen upstream reduct does not entail the complete packet. The result concerns source ownership, not the logically different statement that a realized POVM outcome must have nonzero probability in every state. No stress- or current-difference is claimed because the physical unprobed action and field state are identical. Additional source-owned occupation or an empirically discriminating source-frozen prediction is required. $\square$

This finite counterpair is the stopping result for the origin question at the present source level. It forbids both an unrestricted parent-law promotion of Theorem 1 and another same-level search for a hidden normalization. It does not weaken the conditional forward construction.

Figure 1 summarizes the resulting generative and variational order.

12 Quantum backreaction and the quantum-gravity boundary

The quantum term in Eq. (19) is a renormalized expectation value on a classical body-derived metric, in the standard semiclassical sense [Birrell and Davies, 1982, Wald, 1977]. Local covariance and admissible renormalization follow the usual curved-spacetime QFT framework [Hollands and Wald, 2001, Brunetti et al., 2003]. The metric can also be driven by a stress-noise second cumulant in a stochastic extension, but this does not quantize geometry.

Paper VII therefore establishes a conditional quantum-gravity *interface*, not quantum gravity. A path integral or operator algebra for the metric, gravitational gauge fixing, a physical inner product and a nonperturbative ultraviolet completion remain absent.

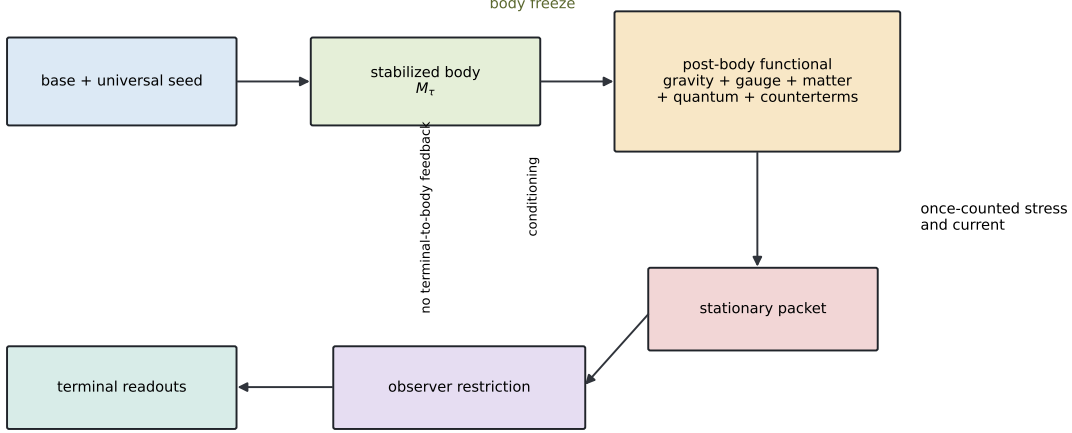


Figure 1: Ordered common-source construction. The body is stabilized before the post-body functional is solved; observer restriction is terminal.

12.1 Two-mode common-preparation theorem

The preceding interface can be made more specific without quantizing the metric. Fix an oriented observer Sachs screen and an occupied two-mode QFT restriction with annihilation operators a_m, a_{m+2} . Let

$$C_{ij} := \langle a_i^\dagger a_j \rangle, \quad N_O := \text{Tr } C > 0, \quad \rho_{\text{ROOT}} := \frac{C}{N_O}. \quad (30)$$

The two screen weights differ by two, so their coherence transforms in the same spin-two representation as the complex screen stress $T_+ + iT_\times$. Denote the once-counted normal-ordered STF overlap of the two occupied modes by $K_{m,m+2}$. It is not a free terminal gain. Let u_i be the source-normalized mode functions, let $e_+^\mu = (e_1^\mu + ie_2^\mu)/\sqrt{2}$ be the oriented complex screen vector, and let f_O be a normalized observer smearing window. If $\mathcal{T}_{\mu\nu}[u_i^*, u_j]$ denotes the polarized bilinear appearing in the once-renormalized Hilbert stress derived from the same QFT action, define

$$K_{ij}[u, f_O] := \int d\mu_O f_O e_+^\mu e_+^\nu \mathcal{T}_{\mu\nu}[u_i^*, u_j]. \quad (31)$$

Thus the action, mode normalization, screen, smearing and subtraction scheme determine K_{ij} . The common-preparation identity is

$$\langle : T_+ + iT_\times : \rangle = K_{m,m+2} C_{m,m+2} = N_O K_{m,m+2} (\rho_{\text{ROOT}})_{m,m+2}. \quad (32)$$

Proposition 8 (Conditional ROOT–TT common preparation). *Assume (i) the physical restriction occupies a common, oriented screen pair whose weights differ by two; (ii) the relevant state is restricted to a number-preserving two-mode second-moment sector, so anomalous moments $\langle a_i a_j \rangle$ and $\langle a_i^\dagger a_j^\dagger \rangle$ vanish or are independently projected out; (iii) all other mode, vacuum and polarization contributions to the selected STF component are orthogonal or included once in the declared subtraction; (iv) the ROOT one-particle reduced state and the stress expectation are formed from the same source-owned QFT preparation; (v) normal ordering or renormalization is entered once in the source ledger; and (vi) $K_{m,m+2} \neq 0$, with $K_{m,m+2}$ calculated by Eq. (31) from the declared*

source preparation, screen and subtraction prescription. Then Eq. (32) follows and, after $K_{m,m+2}$ is source-frozen, admits no additional independent terminal-specific gain within this restricted sector.

Proof. The ROOT one-particle reduced state is the normalized one-particle coherence matrix by Eq. (30). Under assumptions (ii)–(iii), projection of the normal-ordered quadratic stress onto the selected spin-two component leaves $K_{m,m+2}\langle a_m^\dagger a_{m+2} \rangle$. Substitution of $C = N_O \rho_{\text{ROOT}}$ gives Eq. (32). Introducing an additional gain after $K_{m,m+2}$ is fixed would assign two calibrations to one occupied preparation and would violate the once-counted source ledger. The formula does not select the mode pair, screen or subtraction prescription and does not prove that the resulting overlap is nonzero. Changing those source-side inputs changes the calculated $K_{m,m+2}$; it does not authorize a post-hoc terminal gain. $\square$

The use of the unnormalized matrix C , rather than only ρ_{ROOT} , also closes the quantum-induced part of the P1 gluing. Let $J_C : \delta s \mapsto \delta C$, let π_{12} extract the real and imaginary parts of $C_{m,m+2}$, let M_K denote real multiplication by the source-frozen complex number $K_{m,m+2}$, and let L_{P1}^{TT} be the retarded linear response from the screen-TT stress plane to the P1 geometry rows. Then

$$\boxed{J_{P1}^{(Q)} = L_{P1}^{\text{TT}} M_K \pi_{12} J_C}, \quad \ker J_C \subseteq \ker J_{P1}^{(Q)}. \quad (33)$$

If $K_{m,m+2} \neq 0$ and L_{P1}^{TT} is injective on the occupied TT plane, this restriction has real rank two. This is a factorization theorem for the quantum-induced stress row, not for the complete P1 terminal. The distinction is essential: $J_Q(x, y, z) = (x, y)$ and $J_{P1}(x, y, z) = (x, y, z)$ have a rank-two common submap although $(0, 0, 1) \in \ker J_Q \setminus \ker J_{P1}$. Thus local ROOT–TT rank two does not prove the full-carrier zero-cokernel condition. Conversely, replacing C by normalized ρ_{ROOT} alone loses this factorization under occupation-scale variations: $\delta C = \epsilon C$ gives $\delta \rho_{\text{ROOT}} = 0$ but a nonzero TT-stress variation whenever the coherence and overlap are nonzero.

On the physical pure-state ROOT tangent the existing body-first staging makes this statement stronger. With observer and radiative boundary data frozen, the direct Green row, stabilized body stress and direct ROOT-to-occupied-matter term have zero variation. Therefore

$$\boxed{J_{P1}|_{\mathcal{V}_Q} = J_{P1}^{(Q)}|_{\mathcal{V}_Q} = L_{P1}^{\text{TT}} M_K \pi_{12} J_C|_{\mathcal{V}_Q}}. \quad (34)$$

Thus Eq. (33) is the full local P1 factorization on the two-real-dimensional post-body ROOT tangent, while the counterexample above continues to exclude a global conclusion on the complete source domain.

The local result can nevertheless be removed exactly from the remaining global rank calculation. Choose the source-metric orthogonal splitting $\mathcal{E}_{\text{phys}} = \mathcal{E}_R \oplus \mathcal{V}_Q$ and write $J_Q = (C_R, D)$ and $J_{P1} = (A_R, \iota D)$, where D is injective on the two-dimensional ROOT tangent and $\iota = L_{P1}^{\text{TT}} M_K \pi_{12}$. If P_D projects onto $\text{im } D$, define

$$\mathcal{R}_{\text{rest}} x = ((I - P_D)C_R x, (A_R - \iota C_R)x), \quad x \in \mathcal{E}_R. \quad (35)$$

Then projection to $\mathcal{E}_R$ gives

$$\boxed{\ker \begin{pmatrix} J_Q \\ J_{P1} \end{pmatrix} \simeq \ker \mathcal{R}_{\text{rest}}, \quad \text{rank} \begin{pmatrix} J_Q \\ J_{P1} \end{pmatrix} = 2 + \text{rank } \mathcal{R}_{\text{rest}}}. \quad (36)$$

Indeed, $C_R x + D y = 0$ is soluble exactly when $(I - P_D)C_R x = 0$ and then has the unique solution $y = -D^{-1}C_R x$; the P1 equation becomes $(A_R - \iota C_R)x = 0$. Thus the already closed

ROOT tangent contributes exactly two ranks. The unresolved differential content is confined to the rest-domain transverse quantum component and any independent P1 component. This is an exact staged reduction, not physical realization of the complete base-seed packet or a replacement for finite-support and algebraic faithfulness tests.

Nor do the current marginal data determine rank $\mathcal{R}_{\text{rest}}$. Take $\mathcal{E}_R = \mathbb{R}^4$, $D = \iota = I_2$, $C_R = (I_2 \ 0)$ and respectively

$$A_0 = (I_2 \ 0), \quad A_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \quad A_2 = (0 \ I_2). \quad (37)$$

All three controls have the same faithful local ROOT block, raw ranks and marginal Grams $J_Q J_Q^\dagger = J_{P1,k} J_{P1,k}^\dagger = 2I_2$, but rank $\mathcal{R}_{\text{rest},k} = k$. Thus the remaining rank requires the rest-domain mixed P1-quantum incidence. It is calculated by the common-source mixed-Gram formula once the full packet is physically selected; separate marginal closure does not select it.

The inherited common rank-four body theorem removes a larger local block. Let the local source split be $X_B \oplus H_{\text{hid}}$ and write the retained rank-four observer descriptor and P1 differential as

$$J_4 = (A \ B), \quad J_{P1} = (P \ Q), \quad A : X_B \xrightarrow{\sim} Y_4. \quad (38)$$

Full local qubit and Lorentz retention makes A invertible. Defining

$$\Delta_{\text{hid}} = Q - P A^{-1} B \quad (39)$$

and applying exact block elimination gives

$$\boxed{\ker \begin{pmatrix} J_4 \\ J_{P1} \end{pmatrix} \simeq \ker \Delta_{\text{hid}}, \quad \text{rank} \begin{pmatrix} J_4 \\ J_{P1} \end{pmatrix} = 4 + \text{rank} \Delta_{\text{hid}}.} \quad (40)$$

Hence the P1 map factors uniquely through the observer/ROOT descriptor on X_B ; no selector is missing on that common core. If the descent is body-block preserving, then $B = 0$ and the remaining defect is simply Q . This result does not determine the hidden-fibre block, identify the local rank-four packet with the complete 64-real-dimensional P3 descriptor, or prove physical occupation of either completion.

The complete P3 comparison can be reduced on the same source split. Write $J_{P3} = (R \ S)$ and remove its already counted common-core component:

$$\Gamma_{\text{hid}} := S - R A^{-1} B. \quad (41)$$

The same invertible block operations give

$$\text{rank} \begin{pmatrix} J_4 \\ J_{P3} \\ J_{P1} \end{pmatrix} = 4 + \text{rank} \begin{pmatrix} \Gamma_{\text{hid}} \\ \Delta_{\text{hid}} \end{pmatrix}. \quad (42)$$

Consequently the hidden P1 defect is represented by P3 exactly when

$$\boxed{\ker \Gamma_{\text{hid}} \subseteq \ker \Delta_{\text{hid}}} \iff \Delta_{\text{hid}} = L_{P1 \leftarrow P3} \Gamma_{\text{hid}} \quad \text{on } \text{im } \Gamma_{\text{hid}}. \quad (43)$$

This is a source-fibre condition, not a target-rank condition. Indeed, $\Gamma_{\text{hid}}(x, y) = x$ is onto its one-dimensional P3 target while $\Delta_{\text{hid}}(x, y) = y$ detects a P3-null source direction. Thus complete P3 operator-state separation does not by itself close the P1-P3 gluing. Source-fibre injectivity of

Γ_{hid} would be sufficient, while the weaker kernel inclusion in Eq. (43) is necessary and sufficient. Neither condition is presently selected by the physical base-seed law.

The remaining branch choice is represented by one canonical source-fibre obstruction,

$$\mathcal{O}_{P1|P3} := \Delta_{\text{hid}}|_{\ker \Gamma_{\text{hid}}}. \quad (44)$$

Rank-nullity and Eq. (42) give the exact novelty identity

$$\boxed{\text{rank} \begin{pmatrix} J_4 \\ J_{P3} \\ J_{P1} \end{pmatrix} - \text{rank} \begin{pmatrix} J_4 \\ J_{P3} \end{pmatrix} = \text{rank} \mathcal{O}_{P1|P3}.} \quad (45)$$

Thus $\mathcal{O}_{P1|P3} = 0$ is equivalent to complete P1 factorization through P3 after the common core has been counted. If the obstruction is nonzero, its source direction is P1-visible, so it is neither a joint readout null nor an unloaded Schur-neutral presentation auxiliary; source-minimal quotienting cannot remove it. The current parent laws admit both branches and do not select the value of this obstruction.

The common source Hessian also gives a direct evaluation formula. Let $K_{\text{hid}} > 0$ be its Schur complement after the common rank-four block is eliminated and define

$$\hat{\Gamma} = \Gamma_{\text{hid}} K_{\text{hid}}^{-1/2}, \quad \hat{\Delta} = \Delta_{\text{hid}} K_{\text{hid}}^{-1/2}, \quad \Pi_{\ker \hat{\Gamma}} = I - \hat{\Gamma}^+ \hat{\Gamma}. \quad (46)$$

Then

$$\boxed{\text{rank} \mathcal{O}_{P1|P3} = \text{rank} (\hat{\Delta} \Pi_{\ker \hat{\Gamma}}).} \quad (47)$$

Equivalently, form the marginal and mixed Grams of $\hat{\Gamma}$ and $\hat{\Delta}$. If $\mathcal{C}_{\Gamma\Delta}$ is their support-normalized mixed Gram, the principal-angle theorem gives

$$\text{rank} \mathcal{O}_{P1|P3} = \text{rank} \Delta_{\text{hid}} - \#\{j : \sigma_j(\mathcal{C}_{\Gamma\Delta}) = 1\}. \quad (48)$$

Thus one physically selected common incidence packet computes the obstruction without an additional terminal coefficient. The Hessian alone does not fix it: distinct typed incidences over the same positive metric can realize zero or nonzero obstruction.

This obstruction also separates two notions of completeness that must not be conflated. Let $N_{\text{phys}} \subset H_{\text{hid}}$ be the source null annihilated by every retained terminal, so in particular $N_{\text{phys}} \subseteq \ker \Delta_{\text{hid}}$. If P3 is complete on physical source fibres modulo this joint null, then

$$\ker \Gamma_{\text{hid}} \subseteq N_{\text{phys}} \subseteq \ker \Delta_{\text{hid}}, \quad (49)$$

and therefore

$$\boxed{\text{P3 physical-source completeness} \implies \mathcal{O}_{P1|P3} = 0 \implies J_{P1} = L_{P1 \leftarrow P3} J_{P3}} \quad (50)$$

after the common core and joint null have been removed. Conversely, a nonzero obstruction provides a P3-null source direction that P1 still sees; it therefore certifies failure of P3 source-fibre completeness even when P3 is onto and informationally complete on its quantum operator-state target. Equation (50) is a conditional completeness criterion, not a proof that the physical base-seed parent selects the source-complete branch.

The existing seed-chain/Hodge construction fixes a genuine subpacket but does not close this selection problem. It supplies a nonzero nilpotent incidence on the occupied chain image. On a source split $E_{\text{phys}} = E_{\text{chain}} \oplus H_{\text{ext}}$, however, fix the same positive metric and take

$$\Gamma_{\text{hid}} = (1, 0, 0), \quad \Delta_0 = (2, 0, 0), \quad \Delta_1 = (2, 1, 0). \quad (51)$$

The two completions have the same nonzero chain restriction and the same onto P3 map, while $\Delta_0|_{\ker \Gamma_{\text{hid}}} = 0$ and $\text{rank}(\Delta_1|_{\ker \Gamma_{\text{hid}}}) = 1$. Thus the proved chain incidence must be retained and may not be fitted again, but it does not select the complete terminal-typed extension J_* . That remaining extension is the physical representation of the primitive relation presentation on H_{ext} , not another Hodge coefficient or terminal gain.

One selected chain block would extend uniquely if a source-owned typed-isometric transport groupoid made its orbit span E_{P3} and respected all stabilizer relations. This sufficient route is unavailable in the current typed ledger: the two-role observer–instrument defect and the three-role photon–matter defect cannot belong to one type-preserving source orbit. Nor does cyclicity of the represented P3–record GNS carrier replace this condition. For example, the defining $M_2(\mathbb{C})$ representation is cyclic from e_1 because $M_2(\mathbb{C})e_1 = \mathbb{C}^2$, whereas a typed diagonal source group preserves the line $\mathbb{C}e_1$. Thus target-algebra cyclicity does not imply typed source-orbit generation. At least one source-owned inter-orbit relation or separately occupied orbit representative is still required for the full extension.

This remaining relation has an exact finite closure criterion. Decompose the typed source carrier into its source orbits, $V_r = \bigoplus_{\alpha} V_{\alpha}$, let $\mathcal{A}_0$ be the star-algebra generated by the fixed intra-orbit representations, and let B contain the proposed source-owned inter-orbit bridge blocks. Define

$$\mathcal{A}_{\text{br}} = \text{alg}^*(\mathcal{A}_0, B). \quad (52)$$

For fixed orbit metrics, two relative gluings preserve all generated source relations exactly when their relative unitary lies in $U(\mathcal{A}'_{\text{br}})$. Assuming $G_{\text{decl}} \subseteq U(\mathcal{A}'_{\text{br}})$, the gluing is unique modulo the declared gauge group precisely when

$$\boxed{U(\mathcal{A}'_{\text{br}}) = G_{\text{decl}}}. \quad (53)$$

This is an internal sufficiency-and-uniqueness theorem, not an existence theorem for B . In the two-scalar-orbit control, the two orbit projectors alone have a two-dimensional commutant and leave an independent relative phase. Adding one nonzero off-diagonal bridge and its dagger generates $M_2(\mathbb{C})$ and reduces the commutant to the common scalar line. This bridge is required only when the declared gauge is the common scalar. Under the inclusion assumed above, the actual obstruction is the excess commutant

$$d_{\text{excess}} = \dim U(\mathcal{A}'_{\text{br}}) - \dim G_{\text{decl}}. \quad (54)$$

If the two diagonal phases are instead the declared two-sector gauge-superselection center, the unbridged model already has $d_{\text{excess}} = 0$; an off-diagonal bridge would then erase a declared superselection direction and is inadmissible without an upstream refinement of the gauge quotient.

This qualification does not close the P3 source problem. The existing superselection theorem concerns the quark/lepton multiplet split $\{Q, u^c, d^c\}$ versus $\{L, e^c\}$, whereas the present no-go separates relation defects with typed supports $\{Q, O\}$ and $\{R, M, G\}$. These are not the same decomposition. Moreover, Q denotes the ROOT/quantum role in the P3 packet but the quark-doublet module in the multiplet theorem; equality of the glyph is not a typed intertwiner. No current theorem transfers the former gauge center to the latter defect carrier. The current base–seed packet therefore fixes neither the relevant G_{decl} nor, if a positive excess remains, the required

bridge. The adopted P3 handoff is post-body data and cannot be used backwards to generate this upstream source relation.

More precisely, a target gauge action U_g on a carrier H induces a gauge action on the relation-defect carrier V_r only after a source-owned injective intertwiner $\iota_r : V_r \rightarrow H$ is supplied and $U_g \text{im } \iota_r \subseteq \text{im } \iota_r$. Then and only then the pullback is unique:

$$\iota_r \gamma_r(g) = U_g \iota_r, \quad G_{\text{decl}} = \gamma_r(G_{\text{phys}}) / \ker \gamma_r. \quad (55)$$

The odd-Clifford central projectors act on $\Lambda^\bullet E_{P3}$, the two-point record projectors on R_2 , and the quark/lepton center on multiplet modules. None acts on V_r through a currently derived ι_r . Their common two-sector count and equal trace weights therefore do not determine G_{decl} on the typed defect orbits.

There is nevertheless a source-internal positive route. The primitive faithful GNS completion already contains a typed injection $\iota_r : V_r \rightarrow \mathcal{A}_{\text{src}}/N_{\omega_s}$. Let $G_{\text{GNS}}^{\text{typed}}$ be its state-preserving star-automorphisms that preserve the image and role typing. Their GNS implementation is unitary, so the gauge-null subgroup relevant to the gluing theorem is determined by

$$\boxed{G_{\text{decl}}^{\text{GNS}} = \gamma_r(G_{\text{GNS}}^{\text{typed}}) \cap U(\mathcal{A}'_{\text{br}}).} \quad (56)$$

Consequently no independent gauge quotient is needed after that complete GNS source packet is selected. What remains open is precisely its physical base-seed selection and whether its typed stabilizer exhausts the physical gauge group.

The canonical exterior-record construction does not prove that selection. For its already supplied positive role metric K_* , the creation-contraction operator $C_{K_*}(u) = \epsilon(u) + \iota_{K_*(u, -)}$ obeys the Clifford anticommutator. Cyclicity of the normalized exterior trace therefore gives

$$\boxed{\text{Re } \tau_{\text{ext}}(C_{K_*}(u)^* C_{K_*}(v)) = K_*(u, v).} \quad (57)$$

This proves exact compatibility and the absence of a free GNS metric gain, but it does not derive K_* : the contraction leg already contains that metric. A reverse use of Eq. (57) would be circular. The noncircular source target is instead a premetric typed presentation

$$g_r : V_r \longrightarrow \mathcal{A}_{P3}, \quad j_0 : \mathcal{A}_{P3} \hookrightarrow \mathcal{C}_{16}, \quad (58)$$

fixed without K_* . If j_0 is unital and injective and $j_0 g_r$ is injective on the physical defect quotient, then

$$G_0(u, v) = \text{Re } \tau_{16}[(j_0 g_r(u))^* (j_0 g_r(v))] \quad (59)$$

is positive definite. The once-counted relation action can then derive its Hessian, GNS carrier, trace weights and conditional gauge pullback from this one packet. The present manuscript proves the implication, not physical base-seed occupation of Eq. (58).

The frozen P3 Clifford candidate gives one noncircular realization of this target. Take the typed basis (B, O, R, U, Q, M, G) , let g_r map it to the seven self-adjoint generators of $\text{Cl}_7(\mathbb{C})$, and let j_0 be their faithful normalized Jordan-Wigner embedding as the first seven generators of $\text{Cl}_{16}(\mathbb{C})$. Their defining anticommutator and the normalized trace give

$$\text{Re } \tau_{16}[(j_0 g_r(u))^* (j_0 g_r(v))] = \langle u, v \rangle_{\mathbb{R}^7}. \quad (60)$$

Hence this declared completion jointly supplies the faithful GNS packet, identity defect metric, equal odd-Clifford central weights and the conditional gauge pullback without importing a Hessian. It

proves nonemptiness, not unrestricted parent-law realization. Unequal rescalings $z_a \mapsto \lambda_a z_a$ retain the envelope, dagger and role labels while changing the trace metric to $\text{diag}(\lambda_a^2)$. The current base–seed laws contain no typed full-role operator-square incidence that selects the normalized Clifford relation over this counterfamily.

There is, however, a strictly weaker conditional selector than a transitive symmetry among the physically different roles. Suppose all seven typed P3 lines have a source-owned degree-one lift into one primitive exterior calculus. Let ϵ_a and ι_a be its creation and contraction maps, with

$$\{\epsilon_a, \epsilon_b\} = 0, \quad \{\iota_a, \iota_b\} = 0, \quad \{\iota_a, \epsilon_b\} = \delta_{ab}I, \quad \epsilon_a^\dagger = \iota_a. \quad (61)$$

If the self-adjoint response is role-local and has no independent degree-one bypass, then $C_a = \lambda_a(\epsilon_a + \iota_a)$. If, in addition, every occupied primitive slot is counted once in the same source action unit,

$$\tau(C_a^2) = 1 \quad \text{for every } a, \quad (62)$$

then $\lambda_a^2 = 1$ and Eq. (61) gives

$$\boxed{\{C_a, C_b\} = 2\delta_{ab}I.} \quad (63)$$

The remaining signs are primitive-frame reflections and do not alter the metric or Clifford algebra. Thus common primitive cost removes the unequal role-local rescalings without identifying the terminal roles. Role-locality is essential: two unit-cost generators related by an oblique mixture can have a nonzero cross anticommutator.

This reduces but does not close unrestricted parent-law realization and occupation. The existing PUSC3 law owns exterior closure and common unit cost on its declared finite local primitive source algebra. The current base–seed derivation does not yet construct a complete typed lift of (B, O, R, U, Q, M, G) into that one primitive slot class or exclude a separate degree-one bypass on its image. The open physical statement is therefore this typed primitive lift, rather than an arbitrary metric coefficient or a new target carrier.

Nor can the closest existing off-diagonal constructions be substituted by matrix shape alone. The rooted $R_{\text{root}}/\text{TC-CVP1}$ Hessian acts between pair-difference and pair-common W_2 occurrence modules, not between the P3 defect orbits. The wall–observer/internal separator candidate leaves distinct multiplet blocks disconnected, while the existing EACER, seed-normal and BCCT squared-Hodge carriers have no typed full-role incidence from E_{P3} . These are domain failures, not missing numerical gains.

The later P3 reconstruction reduces the premise without changing this no-go. On the declared enriched completion, the canonical exterior–record realization fixes the P3 matrix carrier, its joint record occupation, the minimal cyclic/GNS representation and the central trace weights. Contractible unloaded Schur-neutral presentation directions are quotient-equivalent. These facts remove four apparent selectors, but they begin after the typed positive P3 quotient and its metric have been supplied. They do not derive the base–seed tangent map J_* , its P1 and observer restrictions, the exact closure of a loaded or typed-visible complementary internal sector, or an isolated rank-two ROOT cluster of the full observer-shortened Hessian. Thus the remaining source packet is finite and explicit:

$$\boxed{\begin{array}{l} J_*, \quad \text{exact complement closure,} \quad \text{gap}_{\text{ROOT}}(K_{*,O}) > 0, \\ \text{physical realization of the enriched completion} \end{array}} \quad (64)$$

Once this packet is source-owned, Eq. (91) evaluates the mixed incidence and Eq. (36) evaluates the stacked rank. A new P3 matrix realization, gain or terminal-specific selector cannot supply the missing value.

In particular, the later quantum factor-designation result does not close this source problem. Its observer-event incidence is defined only on an already adopted quantum event quotient: under exact recovery it selects one logical M_2 embedding, after which canonical descent fixes the complementary M_4 realization algebra. That map is neither J_* nor the complete body–observer incidence into the joint P1–P3 recovery ideal. The domains and codomains differ, so factor designation cannot imply the missing rest-domain rank or full typed surjectivity.

The mode-pair premise and the nonzero-overlap premise admit a sharper finite certificate. Let P_O be the occupied rank-two projector on the selected one-particle sector and let J_{scr} be the self-adjoint generator of rotations of the oriented observer screen. In a J_{scr} eigenbasis write

$$P_O C P_O = \begin{pmatrix} p_m & c \\ c^* & p_{m+2} \end{pmatrix}. \quad (65)$$

Proposition 9 (Finite ROOT–TT activation certificate). *Within the declared number-preserving quadratic matter sector, the selected rank-two cluster is a covariantly closed source of a nonzero two-component TT mean of real rank two if and only if*

$$[P_O, J_{\text{scr}}] = 0, \quad \text{spec}(P_O J_{\text{scr}} P_O) = \{m, m+2\}, \quad c \neq 0, \quad K_{m,m+2} \neq 0. \quad (66)$$

For the opposite helicity the spectral gap is -2 .

Proof. Global phase invariance leaves the quadratic coherences $C_{mn} = \langle a_m^\dagger a_n \rangle$. Under a screen rotation they carry character $n - m$, whereas $T_+ + iT_\times$ carries character $+2$. Orthogonality of $SO(2)$ characters therefore removes every off-diagonal coherence except $n - m = 2$ (or -2 for the conjugate helicity). Projector commutation makes these weights well defined inside the occupied cluster. Equation (32) then reduces the TT mean to the complex product $K_{m,m+2}c$. Multiplication by a nonzero complex number has real matrix $\begin{pmatrix} \Re K & -\Im K \\ \Im K & \Re K \end{pmatrix}$, whose determinant is $|K|^2$; hence the response has real rank two exactly when both factors are nonzero. If projector invariance fails, the selected subspace is not a covariantly closed two-mode source. Within an invariant cluster, failure of the weight, coherence or overlap condition makes the selected quadratic TT mean vanish. $\square$

Four controls delimit the statement. First, a single geometric-optics null ray has stress $Ik \otimes k$ and zero STF projection on its own Sachs screen; non-collinearity relative to a common screen is therefore essential. For a ray with transverse components (p_1, p_2) ,

$$T_+ + iT_\times = \frac{I}{2}(p_1 + ip_2)^2, \quad (67)$$

whose real Jacobian has rank two away from $p_1 = p_2 = 0$. Second, ρ_{ROOT} alone cannot determine the stress magnitude: preparations with the same normalized state and different N_O have different stress. Third, an incoherent classical anisotropic stress can be physically relevant, but it does not establish the coherence-controlled identity above. Fourth, outside the declared number-preserving sector the full covariance packet, including anomalous moments, replaces C ; Eq. (32) must not be extrapolated to a generic QFT state.

There is also an explicit nonzero witness for the final condition in Eq. (66). On a patch where an isolated ray has fixed nonzero transverse direction $p_1 + ip_2$ relative to the common screen, positive intensity I and a nonnegative smearing function f_O with nonzero integral, its smeared STF coefficient is proportional to

$$K_{\text{ray}} = \frac{1}{2}(p_1 + ip_2)^2 \int d\mu_O f_O I \neq 0. \quad (68)$$

This proves that the admissible nonzero-overlap class is nonempty. It does not show that an arbitrary multimode packet avoids destructive cancellation, or that the physical parent selects this witness.

Proposition 8 is a conditional common-source bridge, not an unrestricted parent-realization theorem. This paper does not prove that the physical base-seed system occupies the required non-collinear $\Delta m = 2$ pair, that every ROOT carrier has this two-mode realization, or that geometry is quantized.

12.2 Minimal invariant activation potential

The four activation conditions need not be independent switches. Let p be a weight-one complex transverse coordinate on the common screen and c the weight-two relative coherence. Under screen rotations $p \mapsto e^{i\theta}p$ and $c \mapsto e^{2i\theta}c$, the lowest-total-degree analytic real mixed invariant is unique up to one coefficient and complex conjugation, $\text{Re}(c^*p^2)$. Higher-order invariants are not excluded. Consider

$$\mathcal{V}(c, p) = r_c|c|^2 + u_c|c|^4 + r_p|p|^2 + u_p|p|^4 - 2\lambda \text{Re}(c^*p^2), \quad (69)$$

where $u_c, u_p, r_c > 0$, $r_p < 0$ and $\lambda \in \mathbb{R} \setminus \{0\}$.

Proposition 10 (Conditional minimal activation). *Every global minimizer of Eq. (69) has $p_* \neq 0$ and $c_* \neq 0$, with the relative phase locked by $\arg c_* = 2 \arg p_*$ up to the sign of λ . If the once-counted STF overlap is $K(p) = \kappa p^2$ with $\kappa \neq 0$, the full ROOT-TT activation certificate holds.*

Proof. The positive quartics make $\mathcal{V}$ coercive. At $p = 0$, its unique minimum is $c = 0$ with value zero. Along $c = 0$, however, $|p|^2 = -r_p/(2u_p)$ gives $\mathcal{V} = -r_p^2/(4u_p) < 0$, so a global minimizer cannot have $p = 0$. At $p \neq 0$, the c^* variation evaluated at $c = 0$ is $-\lambda p^2 \neq 0$, so $c = 0$ is not stationary. Phase minimization gives the doubled-angle lock. The square p^2 has screen weight two and therefore selects the $\Delta m = 2$ coherence, while $p_* \neq 0$ and $\kappa \neq 0$ give nonzero STF overlap. $\square$

This proposition closes activation only inside the declared minimal completion. The current physical reduct does not determine $r_p < 0$, $\lambda \neq 0$ or $\kappa \neq 0$. In particular, $\lambda = 0$ is an exact same-sector control with $c_* = 0$.

12.3 Source-loaded activation from an occupied Abelian field

The spontaneous branch above is not the only realization. If a nonzero transverse coordinate p_{src} is already supplied by the occupied source ledger, minimize only

$$\mathcal{V}_{p_{\text{src}}}(c) = r_c|c|^2 + u_c|c|^4 - 2\lambda \text{Re}(c^*p_{\text{src}}^2), \quad r_c, u_c > 0. \quad (70)$$

Proposition 11 (Source-loaded coherence selector). *If $p_{\text{src}} \neq 0$ and $\lambda \neq 0$, Eq. (70) has a unique minimizing amplitude and it obeys*

$$(r_c + 2u_c|c_*|^2)c_* = \lambda p_{\text{src}}^2, \quad c_* \neq 0. \quad (71)$$

Its phase has screen weight two. Therefore this branch closes the coherence condition of Proposition 9 without assuming $r_p < 0$.

Proof. For fixed phase, the radial factor on the left side of Eq. (71) is strictly increasing from zero to infinity. It consequently has exactly one positive solution for every nonzero right side, and phase minimization aligns c_* with λp_{src}^2 . Since p_{src} has screen weight one, its square has weight two. $\square$

The mixer in Eq. (70) is also fixed once geometry and stress are varied in the same functional. With the complex STF convention $q_h = h_+ + ih_\times$ and $q_T = T_+ + iT_\times$, write

$$q_h = \beta p_{\text{src}}^2, \quad q_T = Kc. \quad (72)$$

Proposition 12 (Metric-variation ownership of the mixer). *Suppose the matter stress is defined by variation of the same once-counted matter functional used for the ROOT preparation,*

$$\delta_g \Gamma_{\text{matter}} = -\frac{1}{2} \int d^4x \sqrt{-g} T^{\mu\nu} h_{\mu\nu}. \quad (73)$$

On the common screen, Eq. (72) generates a mixed term

$$-g_{\text{TT}} \text{Re}(\beta K^* c^* p_{\text{src}}^2), \quad g_{\text{TT}} > 0, \quad (74)$$

where g_{TT} is fixed by the declared screen and smearing normalization. After the common source phase is fixed,

$$\lambda = \frac{g_{\text{TT}}}{2} |\beta K|. \quad (75)$$

Thus the mixer is nonzero exactly when the geometric TT pullback and the once-counted stress overlap are both nonzero; it is not an independent cross-terminal gain.

Proof. For two real STF screen tensors, their contraction is proportional to $\text{Re}(q_T^* q_h)$. Substitution of Eq. (72) into Eq. (73) gives Eq. (74); the positive proportionality factor contains only the frozen screen, measure and normalization conventions. A common phase choice puts its coefficient in the real form used in Eq. (70), yielding Eq. (75). Adding another multiplier would count the same metric variation twice. $\square$

The Abelian field supplies a canonical candidate for p_{src} . Define

$$I_F = \frac{1}{2} F_{ab} F^{ab}, \quad J_F = \frac{1}{2} F_{ab} (*F)^{ab}, \quad \Delta_F = I_F^2 + J_F^2. \quad (76)$$

For $\Delta_F > 0$, the non-null Faraday field has an unordered pair of distinct principal null directions $\{[k_+], [k_-]\}$ [Wylleman et al., 2020]. Given observer u , line-of-sight $q = u + n$ and its oriented screen projector $S_{(u,q)}$, set

$$p_\pm = \frac{S_{(u,q)} k_\pm}{-u \cdot k_\pm}, \quad \chi_{\text{dir}} = |\varepsilon_S(p_+, p_-)|. \quad (77)$$

These expressions are invariant under rescaling or exchange of the null representatives.

Proposition 13 (Conditional Faraday directional selector). *On the non-null Abelian branch, $\chi_{\text{dir}} > 0$ supplies two linearly independent common-screen directions and hence a nonzero transverse source for the loaded coherence selector. For a non-principal observer, writing $k_\pm/(-u \cdot k_\pm) = u + s_\pm$ gives*

$$\chi_{\text{dir}} = |n \cdot (s_+ \times s_-)|, \quad (78)$$

so the directional condition fails only for sight lines on the great circle spanned by s_+ and s_- . A principal observer, a null field $\Delta_F = 0$, and an aligned sight line are exact zero controls.

Proof. The non-null electromagnetic classification gives two distinct principal null lines. Screen projection and normalization produce Eq. (77); its two-dimensional determinant is nonzero exactly for linearly independent projections. Direct projection into $n^\perp$ gives the scalar triple product Eq. (78). For a non-principal observer $s_+ \times s_- \neq 0$, and its orthogonal sight lines form one great circle. For a principal observer $s_- = -s_+$, while a null field has one repeated principal null line, so both controls give zero. $\square$

This removes an arbitrary directional-pair choice *after* occupation of the non-null Abelian branch. It does not prove that the Tau parent selects that branch, that the QFT preparation lift identifies its support with the two-mode ROOT cluster, or that every observer lies outside the exact control sets.

12.4 Faithful common-completion occupation

The hub's common-completion criterion can be stated directly in the present ledger. Let $K_* > 0$ be the true-null-quotiented body Hessian and let $R_{\text{rec}}^{\text{dedup}}$ be the de-duplicated primitive recovery incidence into the required finite recovery ideal. Define

$$G_*^{\text{rec}} = R_{\text{rec}}^{\text{dedup}} K_*^{-1} (R_{\text{rec}}^{\text{dedup}})^\dagger, \quad \rho_*^{\text{rec}} = \frac{G_*^{\text{rec}}}{\text{Tr } G_*^{\text{rec}}}. \quad (79)$$

Proposition 14 (Conditional faithful occupation reduction). *If $R_{\text{rec}}^{\text{dedup}}$ is surjective, then ρ_*^{rec} is faithful on the required recovery ideal and its cyclic GNS representation is faithful and unique up to unitary equivalence. In particular, for every nonzero positive central projector P_A of the required Abelian ideal,*

$$w_A := \text{Tr}(\rho_*^{\text{rec}} P_A) > 0. \quad (80)$$

Thus Abelian block occupation requires no independent terminal occupation bit. If a stable ROOT projector P_O has nonzero weight in the same recovered QFT preparation Γ_O , then

$$\rho_{\text{ROOT}} = \frac{P_O \Gamma_O P_O}{\text{Tr}(P_O \Gamma_O P_O)} \quad (81)$$

and the ROOT state and Hilbert stress are automatically restrictions of one preparation.

Proof. Surjectivity makes G_*^{rec} positive definite on the recovery ideal, so its normalized state is faithful. A faithful state is strictly positive on every nonzero positive element, giving Eq. (80). Faithfulness of the state makes the corresponding finite-dimensional GNS representation faithful, while the standard GNS uniqueness theorem fixes its cyclic realization up to a unitary intertwiner. Finally, Eq. (81) is the normalized compression of the same Γ_O whose metric variation defines the stress; hence no second ROOT preparation is introduced. $\square$

This closes sector occupation and common-state ownership *conditional* on one surjective recovered branch. It does not imply $\Delta_F > 0$ at every spacetime point: faithfulness of the Abelian ideal is a support statement, whereas null versus non-null Faraday type is a local state/solution property. The remaining unrestricted physical problem is therefore finite: the enriched parent law must realize a unique stable branch with surjective $R_{\text{rec}}^{\text{dedup}}$, and the selected observer context must meet the non-null and nondegenerate-screen conditions above. The current reduct does not entail that surjectivity; Proposition 7 supplies its deficient control.

12.5 Exact source-rank reduction

The hub also supplies an exact factorization of the recovery incidence. Let $X_{\text{src}}^{\text{lin}}$ contain the body-owned and irreducible observer–source inputs after the true-null quotient, let $E_{\text{req}}^{\text{lin}}$ be the declared linearized recovery carrier, and write

$$J_{\text{phys}}^{\text{lin}} : X_{\text{src}}^{\text{lin}} \longrightarrow E_{\text{MVP}}^{\text{lin}}, \quad R_{\text{rec}}^{\text{lin}} = U_{\text{rec}} D_{\Sigma} J_{\text{phys}}^{\text{lin}}. \quad (82)$$

Here D_{Σ} is the ordered lower-triangular response map and U_{rec} implements source-owned de-duplication.

Proposition 15 (Exact source-rank criterion). *Suppose D_{Σ} and U_{rec} are isomorphisms on their declared physical quotients and $H_{\text{src}} > 0$ is the once-counted source Hessian. Then*

$$\text{rank } R_{\text{rec}}^{\text{lin}} = \text{rank } J_{\text{phys}}^{\text{lin}}, \quad \ker R_{\text{rec}}^{\text{lin}} = \ker J_{\text{phys}}^{\text{lin}}, \quad H_{\text{phys}} = (J_{\text{phys}}^{\text{lin}})^{\dagger} H_{\text{src}} J_{\text{phys}}^{\text{lin}}. \quad (83)$$

Consequently the realized branch is strictly stable and its differential recovery has zero cokernel if and only if $J_{\text{phys}}^{\text{lin}}$ is an isomorphism. Nonzero projection into every typed role is not sufficient.

Proof. Left composition by the two isomorphisms preserves rank and kernel, proving the first two identities. Moreover $\langle x, H_{\text{phys}} x \rangle = \langle J_{\text{phys}}^{\text{lin}} x, H_{\text{src}} J_{\text{phys}}^{\text{lin}} x \rangle$, so positivity of H_{src} makes strict stability equivalent to injectivity of $J_{\text{phys}}^{\text{lin}}$. Surjectivity of the linearized recovery map is equivalent to surjectivity of $J_{\text{phys}}^{\text{lin}}$. Both properties hold exactly when the latter map is an isomorphism. $\square$

A full-column-rank map from a lower-dimensional source quotient may have a nonzero component in every required role and yield $H_{\text{phys}} > 0$, while still having nonzero cokernel. This is the exact stable-but-incomplete control. Conversely, the hub’s occupied-P3 result supplies a nonempty faithful ROOT-edge subrepresentation inside the declared enriched MVP, but a faithful subrepresentation is not global surjectivity of $J_{\text{phys}}^{\text{lin}}$. Thus the open physical statement is no longer a list of terminal choices: it is whether the enriched base–seed/body plus irreducible observer–source law selects one complete differential incidence with zero kernel and zero cokernel, while the same packet covers the finite record support and acts faithfully on the physical algebra quotient.

Write

$$J_{\text{phys}}^{\text{lin}} = [J_{BS}, J_{OS}^{\text{irr}}], \quad P_{BS} = J_{BS} J_{BS}^+, \quad C_{OS} = (I - P_{BS}) J_{OS}^{\text{irr}}. \quad (84)$$

Proposition 16 (Observer–source complement rank). *On a source-frozen metric chart,*

$$\text{rank } J_{\text{phys}}^{\text{lin}} = \text{rank } J_{BS} + \text{rank } C_{OS}. \quad (85)$$

If $N_{\text{lin}} = \dim E_{\text{MVP}}^{\text{lin}}$, differential recovery has zero cokernel exactly when

$$\text{rank } C_{OS} = N_{\text{lin}} - \text{rank } J_{BS}. \quad (86)$$

Proof. Split the target orthogonally as $\text{im } J_{BS}$ and its orthogonal complement. Column operations remove the component of J_{OS}^{irr} inside the first summand; its remaining component is C_{OS} and has orthogonal image. The ranks therefore add, and Eq. (86) is the zero-cokernel condition. $\square$

The hub’s WR-T113/WR-T117 construction evaluates this complement on the retained six-row PBSE subsystem. Its base plus four access rows have rank five, while the irreducible observer–source row is outside their span and raises the rank to six. Thus, whenever this subsystem embeds faithfully into the Paper-VII linearized carrier,

$$\text{rank } C_{OS} \geq 1. \quad (87)$$

This rules out a purely base–seed factorization on that retained subsystem. It does not determine N_{lin} or prove Eq. (86) for the complete carrier. Moreover differential rank cannot replace finite record coverage or algebraic faithfulness; the three conditions are jointly required for full typed recovery.

12.6 De-duplicated primitive recovery dimension

The broad P1 incidence R_{P1} already includes the common body–matter and causal/time lanes, together with metric/gravity, radiation and observer–screen differentials. On the source-minimal nonzero P1 branch its rank is four. Let R_Q be the real linearization of the occupied two-mode ROOT incidence on the same frozen source quotient and define the source-space row projector

$$P_{P1} = R_{P1}^+ R_{P1}. \quad (88)$$

Proposition 17 (De-duplicated P1–ROOT rank). *For the adopted P1–ROOT–gauge packet,*

$$R_{\text{rec}}^{\text{lin}} = \begin{pmatrix} R_{P1} \\ R_Q(I - P_{P1}) \end{pmatrix}, \quad \text{rank } R_{\text{rec}}^{\text{lin}} = 4 + r_{Q\perp}, \quad (89)$$

where

$$r_{Q\perp} = \text{rank}_{\mathbb{R}} R_Q(I - P_{P1}) = \text{rank}_{\mathbb{R}} R_Q - \dim_{\mathbb{R}}(\text{row } R_Q \cap \text{row } R_{P1}), \quad 0 \leq r_{Q\perp} \leq 4. \quad (90)$$

An invertible gauge transformation on the retained charged/polarization carrier changes the presentation but adds no recovery rank.

Proof. The second block in Eq. (89) lies in the orthogonal complement of the row space of R_{P1} , so the two ranks add. Restricting the orthogonal projection $I - P_{P1}$ to $\text{row } R_Q$ gives Eq. (90) by rank–nullity. The real dimension of the two-mode complex ROOT packet is at most four. Left composition by an invertible gauge map preserves both row-space dimension and stacked rank. $\square$

Separate marginal P1 and ROOT metrics do not determine $r_{Q\perp}$: equal identity marginal Grams admit row-space intersections of real dimension four, two or zero, and hence increments 0, 2, 4. The missing datum is not another gain but the common-source mixed Gram. With a positive source Hessian K_* , observer access A_O , isolated occupied ROOT projector P_Q and faithful ROOT restriction C_Q , it is

$$G_{P1,Q} = R_{P1} K_*^{-1} A_O^\dagger P_Q C_Q^\dagger. \quad (91)$$

After normalizing by the two marginal Grams, the singular values are the cosines of the principal angles between the whitened row spaces. Therefore

$$r_{Q\perp} = \text{rank } R_Q - \#\{j : \sigma_j(G_{P1}^{-1/2} G_{P1,Q} G_Q^{-1/2}) = 1\}. \quad (92)$$

Thus the differential recovery rank is confined to $4 \leq \text{rank } R_{\text{rec}}^{\text{lin}} \leq 8$, but no unique value follows until the enriched physical source selects the common packet $(K_*, R_{P1}, A_O, P_Q, C_Q)$. The PBSE transverse row proves genuine observer–source novelty on its retained subsystem; it must not be added again to Eq. (89) unless an independent embedding outside the broad P1 row space is proved.

Proposition 18 (Conditional seed-presentation selector). *Let the universal seed own a complete finite involutive presentation $\{r_\alpha\}$ of the de-duplicated primitive role algebra and let*

$$\mathcal{A}_{\text{rep}}(j) = \frac{A_*}{2} \sum_{\alpha} \|r_\alpha(j)\|_{G_\alpha}^2. \quad (93)$$

Assume: (i) its zero set is one typed-unitary orbit represented by Π_{phys} ; (ii) the relation derivative is injective on the gauge/true-null quotient, so $K_* = D_r^* G D_r > 0$; (iii) R_{P1} and A_O are fixed typed projections of the same represented tangent $J_* = D\mathcal{E}_{\Pi_{\text{phys}}}$; and (iv) the shorted operator

$$K_{*,O} = (A_O K_*^{-1} A_O^*)^{-1} \quad (94)$$

has an isolated stable occupied complex rank-two cluster with Riesz projector P_Q and faithful represented ROOT restriction C_Q . Then, up to the same typed-unitary gauge, the seed representation selects

$$(K_*, R_{P1}, A_O, P_Q, C_Q), \quad R_Q = C_Q P_Q A_O, \quad (95)$$

and Eqs. (91)–(92) follow without an additional terminal coefficient.

Proof. The single zero orbit fixes Π_{phys} up to typed unitary equivalence. Differentiation of its represented generator evaluation and the fixed role projections give R_{P1} and A_O , while the Hessian of Eq. (93) at the zero gives K_* . Functional calculus of the isolated cluster fixes P_Q covariantly, and faithfulness fixes C_Q up to a terminal frame. Such a frame cancels from the ranks and mixed-Gram principal angles, proving Eq. (95) and the claim. $\square$

This reduces five apparent inputs to two physical premises, but does not prove them. The current seed grammar does not yet supply a complete uniquely represented P3 (or full eight-role) involutive presentation, and the physical observer-shortened Hessian has not been proved to contain the required occupied isolated cluster. An occupied complete ROOT edge alone is insufficient: its internal restriction can be held fixed while the complementary spectrum is chosen either separated or interleaved. The physical spectral proposition is therefore undetermined, not false.

13 Technical-paper handoffs

Technical Paper VII-B proves, inside its declared finite completion class, the rank-seven support–reconstruction, Tau-local subsystem composition, pointed exterior carrier, operator-square selector and class-minimal enriched P3 realization. Technical Paper VII-C separately owns the scalar-15/type-II source preimage, occupation no-go and conditional activation, common-scale ratio and exact functional-RG evaluation boundary. Neither imported result is a theorem that the unrestricted physical base–seed parent occupies its completion.

Corollary 2 (Conditional three-level series closure). *Assume the common-source occupation and once-counted terminal premises proved in this paper and, independently, the occupied class-minimal P3 realization of Technical Paper VII-B. Then the declared series completion is closed at three separate levels: common source, algebra/representation and terminal response. No blockwise gain is introduced by composing the two packets.*

Proof. The present paper supplies the common source and terminal levels. Technical Paper VII-B supplies the algebra/representation level using the same declared unit and typed source ledger. Composition therefore adds no new normalization. The result remains conditional on both papers’ physical-realization premises. $\square$

14 Decision ledger

Statement	Status	Boundary
One once-counted functional generates effect, metric, gauge and clock-source responses	proved conditionally	body and source packet are frozen first
Einstein–Maxwell–matter equations and Ward/Bianchi identities	recovered inside the standard leaf	action shapes and anomaly freedom are premises
Classical and quantum stress are not double counted	proved by ownership ledger	depends on declared renormalization bookkeeping
ROOT and TT arise from one two-mode preparation	proved in the declared sector	ROOT is a one-particle reduced state, not full tomography
ROOT closes all remaining P1 source directions	not established	exact mixed-Gram/source-rank obstruction remains
Standard-limit agreement proves common Tau provenance	disproved	explicit same-reduct counter-models
Unrestricted parent realizes and occupies the full joint packet	open	not inferable from finite terminal agreement
Quantized geometry or full Standard Model	not derived	outside Paper VII-A

15 Occupied-body input inherited from parent realization

The joint terminal ledger is evaluated only after occupied body support is frozen. Upstream finite-local analysis shows that every positive seed-unloaded split sector has zero response, whereas every stable loaded or cross-coupled direction belongs to generated support. Consequently no inert finite-local extension supplies an additional terminal gain or stress term here. This does not derive the gravity–matter–gauge packet from the unrestricted base–seed law and does not establish ambient exhaustivity.

16 Conclusion

Paper VII-A establishes a conditional common-source architecture for gravity, matter, Abelian gauge response, formal clock response and quantum backreaction, together with the exact bookkeeping and source-rank conditions needed to avoid double counting. Its strongest positive bridge is the ROOT–TT same-preparation theorem; its strongest negative result is that standard-limit agreement and finite terminal closure do not identify common parent provenance.

The original monolithic Paper VII has therefore been separated without discarding information. The P3 algebra and law-to-packet proof now belongs to VII-B, and the scalar-15/RG branch to VII-C. The decisive remaining result is still either a deeper physical base–seed derivation of the jointly occupied source packet or a source-frozen, gain-free cross-terminal prediction not implied by standard physics.

A Self-contained dependency certificate

This appendix separates logical premises from provenance. The conclusions of Paper VII use only the following statements. A reader may treat each imported statement as an explicit assumption and verify every theorem in this paper without consulting another Tau manuscript.

D1: body-first order.

There is a C^2 pre-body action $\mathcal{A}_{\text{pre}}[y; B, s]$ with an occupied stationary point y_* and positive

physical Hessian H_B . The post-body functional is evaluated on the frozen parameter y_* ; post-body source variation is not varied back into the body equation. This is the only body-first premise used in Eqs. (8) and (14).

D2: common typed source domain.

On the selected body and observer support there is one finite source domain $\mathcal{E}_{*,O}$, one access map A_O , typed linear role maps $R_a : \mathcal{E}_{*,O} \rightarrow \mathcal{Y}_a$, and stable resolution maps $Q_O^{(a)}$. No direct-sum decomposition of $\mathcal{E}_{*,O}$ is assumed. Proposition 1 is proved here from exactly this premise.

D3: occupied operation data.

An operation algebra $\mathfrak{A}_{\text{src}}$, occupied vector Ω_s and positive root covariance X_0 are given. Whenever the full-occupation branch is used, Ω_s is cyclic and every required R_a is surjective. Whenever the rooted-path branch is used, $\text{supp } X_0 \subseteq V_s$ and every admitted L_e preserves $V_s = \text{span}(\mathfrak{A}_{\text{src}}\Omega_s)$. Proposition 2 proves the resulting full-occupation or activation statement; physical realization of either premise set is not imported.

D4: fixed quantum preparation.

The CTP boundary datum satisfies $\rho_{\text{in},O} \geq 0$ and $\text{Tr } \rho_{\text{in},O} = 1$. It is held fixed while g , A and Φ are varied. The CTP evolution and observer restriction produce a normalized output state ρ_O . CTP response is obtained by varying the difference source before taking the physical equal-branch limit. No Euler variation with respect to a density matrix is used.

D5: standard Lorentzian leaf.

On the selected leaf the geometric and Abelian functionals satisfy the classification hypotheses of Proposition 5, with a common unit packet. Their Einstein–Hilbert and Maxwell shapes then follow inside that class. The matter functional, measure and renormalization prescription are diffeomorphism and $U(1)$ gauge invariant and anomaly-free for the Ward identities invoked here. This is an assumed standard-limit leaf, not a Tau derivation of those actions.

D6: operational effect source.

Each declared outcome family is a POVM, $E_a \geq 0$ and $\sum_a E_a = I$, and its source coupling is Eq. (9). This premise, together with D4, is sufficient for the normalized Born derivative in Eq. (12); no stronger quantum reconstruction theorem is used.

D7: clock-source boundary.

The clock character ϑ is an admitted external source coordinate of the post-body functional. Paper VII uses only the formal response derivative $\delta W_O^{\text{ren}}/\delta\vartheta$. It neither imports nor proves a duration law, global clock integrability or a Tau-specific clock distortion.

D8: once-counted calibration.

Every displayed sector uses one declared action unit, each occupied contribution and counterterm has one ledger owner, and the overlap $K_{m,m+2}$ is calculated from Eq. (31) and frozen before the ROOT–TT comparison. This bookkeeping premise forbids an additional fitted gain but does not select the common unit, occupied mode pair, screen or subtraction prescription.

Provenance boundary

The eight D-items above are fully stated premises. A reader may verify the conditional mathematics without consulting another Tau manuscript. Their provenance does not promote them into

unrestricted physical parent laws.

References

- Julian Schwinger. Brownian motion of a quantum oscillator. *Journal of Mathematical Physics*, 2:407–432, 1961. doi: 10.1063/1.1703727.
- L. V. Keldysh. Diagram technique for nonequilibrium processes. *Soviet Physics JETP*, 20:1018–1026, 1965.
- David Lovelock. The einstein tensor and its generalizations. *Journal of Mathematical Physics*, 12:498–501, 1971. doi: 10.1063/1.1665613.
- Robert M. Wald. The back reaction effect in particle creation in curved spacetime. *Communications in Mathematical Physics*, 54:1–19, 1977. doi: 10.1007/BF01609833.
- N. D. Birrell and P. C. W. Davies. *Quantum Fields in Curved Space*. Cambridge University Press, 1982. doi: 10.1017/CBO9780511622632.
- Stefan Hollands and Robert M. Wald. Local wick polynomials and time ordered products of quantum fields in curved spacetime. *Communications in Mathematical Physics*, 223:289–326, 2001. doi: 10.1007/s002200100540.
- Bei-Lok Hu and Enric Verdaguer. Stochastic gravity: Theory and applications. *Living Reviews in Relativity*, 11:3, 2008. doi: 10.12942/lrr-2008-3.
- Flaminia Giacomini, Esteban Castro-Ruiz, and Caslav Brukner. Quantum mechanics and the covariance of physical laws in quantum reference frames. *Nature Communications*, 10:494, 2019. doi: 10.1038/s41467-018-08155-0.
- Ted Jacobson. Thermodynamics of spacetime: The einstein equation of state. *Physical Review Letters*, 75:1260–1263, 1995. doi: 10.1103/PhysRevLett.75.1260.
- H. Blaine Lawson and Marie-Louise Michelsohn. *Spin Geometry*, volume 38 of *Princeton Mathematical Series*. Princeton University Press, Princeton, NJ, 1989. ISBN 978-0-691-08542-5.
- Ola Bratteli and Derek W. Robinson. *Operator Algebras and Quantum Statistical Mechanics I: C^* - and W^* -Algebras, Symmetry Groups, Decomposition of States*. Springer, Berlin, 1979. doi: 10.1007/978-3-662-02313-6.
- Jamie Vicary. Categorical formulation of finite-dimensional quantum algebras. *Communications in Mathematical Physics*, 304(3):765–796, 2011. doi: 10.1007/s00220-010-1138-0.
- Bob Coecke, Dusko Pavlovic, and Jamie Vicary. A new description of orthogonal bases. *Mathematical Structures in Computer Science*, 23(3):555–567, 2013. doi: 10.1017/S0960129512000047.
- Romeo Brunetti, Klaus Fredenhagen, and Rainer Verch. The generally covariant locality principle: A new paradigm for local quantum physics. *Communications in Mathematical Physics*, 237:31–68, 2003. doi: 10.1007/s00220-002-0722-0.
- Lode Wylleman, L. Filipe O. Costa, and José Natário. Poynting vector, super-poynting vector, and principal observers in electromagnetism and general relativity. *arXiv e-prints*, page arXiv:2007.15384, 2020.